# Exercise Sheet

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**Exercise 1 (Naor–Yung and ElGamal).** Apply the Naor–Yung transform to the ElGamal encryption scheme. Specify the NY language for ElGamal and assume a NIZK proof system $(\text{Setup}_{NY\text{-}E}, \text{Prove}_{NY\text{-}E}, \text{Vfy}_{NY\text{-}E})$ for this language.

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**Exercise 2 (Naor–Yung with randomness reuse).** We consider a variation of the Naor–Yung transformation that uses the same randomness for both encryptions. This transformation does not achieve IND-CCA1 security. Find an encryption scheme $\Pi$ that is IND-CPA secure, but the Naor-Yung transformation with randomness reuse w.r.t this scheme is insecure, i.e., not even IND-CPA.

$\text{Gen}_{NY\text{-}r}(1^\lambda)$:

$(pk_0, sk_0) \leftarrow \text{Gen}(1^\lambda)$, $(pk_1, sk_1) \leftarrow \text{Gen}(1^\lambda)$, $crs \leftarrow \text{Setup}(1^\lambda)$

$pk := (pk_0, pk_1, crs)$, $sk := sk_0$ ($sk_1$ is erased)

$\text{Enc}_{NY\text{-}r}(pk, m)$:

$r \leftarrow_\$ \mathcal{R}$, $c_0 := \text{Enc}(pk_0, m; r)$, $c_1 := \text{Enc}(pk_1, m; r)$

$\pi \leftarrow \text{Prove}(crs,\ x = (pk_0, pk_1, c_0, c_1),\ w := (m, r))$

output $ct := (c_0, c_1, \pi)$

$\text{Dec}_{NY\text{-}r}(sk, ct)$:

parse $ct = (c_0, c_1, \pi)$; if $\text{Vfy}(crs, (pk_0, pk_1, c_0, c_1), \pi) = 0$: output $\perp$

else output $\text{Dec}(sk_0, c_0)$

*Hint 1*: Take an arbitrary IND-CPA encryption scheme $\Pi'$ and modify it to a scheme $\Pi$ which has the desired property: $\Pi$ is IND-CPA, but NY-r instantiated with $\Pi$ is not even IND-CPA.

*Hint 2*: Look for a way to leak the message from two encryptions with the same randomness.

*Hint 3*: $\Pi$ could behave differently depending on the first bit of the public key.

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**Exercise 3 (Designated-Verifier NIZKs).** Similar to publicly verifiable NIZK proof systems, we consider *designated verifier* NIZK proof systems with the following syntax:

- $\text{Gen}(1^\lambda)$: outputs a key pair $pk, sk$.
- $\text{Prove}(pk, x, w)$: outputs a proof $\pi$.
- $\text{Vfy}(sk, x, \pi)$: deterministic, outputs 1 (accept) or 0 (reject).

In a nutshell, here we omit the crs in favor of a key pair for the verifier. The properties completeness, soundness, and ZK transfer naturally to designated verifier NIZKs with this change.

Consider the following designated verifier NIZK proof system for the DDH language, which was introduced in the lecture $\mathcal{L}_{\text{DH}} := \{(g_1^r, g_2^r) : r \in \mathbb{Z}_p\} \subset \mathbb{G} \times \mathbb{G}$. This is exactly the DDH smooth projective hash from the lecture.

$\text{Gen}(1^\lambda) : k_1, k_2 \leftarrow_\$ \mathbb{Z}_p, \quad h := g_1^{k_1} \cdot g_2^{k_2}, \quad sk := (k_1, k_2), \quad pk := h$

$\text{Prove}(pk, x, w) : \text{Parse } pk = h, w = r, \quad \text{output } \pi := h^r$

$\text{Vfy}(sk, x, \pi) : \text{Parse } sk = (k_1, k_2), x = (u_1, u_2), \quad \text{output } \pi \stackrel{?}{=} u_1^{k_1} \cdot u_2^{k_2}$

Write down the definitions for completeness, soundness, and ZK for *designated verifier* NIZKs and show that the designated verifier NIZK proof system above achieves these.

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